

 For **Server (Production or Heavy OCR Inference)**

 Recommended Specs (Cloud/Bare Metal)

Spec	Value
GPU	NVIDIA L40 / L4 (32GB) or
CPU	8+ vCPUs
RAM	64 GB
SSD	100+ GB
Network	1 Gbps
Provider	Paperspace, AWS, Lambda, OVH
Scale	1 server $\approx$ 15–20 documents/minute
OS	Ubuntu 24.04 LTS Linux

 Recommended Bare-Metal Server Specs

Component	Recommended	Notes
GPU	NVIDIA L40 / A40 / A10 / RTX 6000 ADA	For Qwen2-VL inference (needs ~10–12GB VRAM)
CPU	AMD EPYC 7003 series or Intel Xeon Gold (12+ cores)	Needed for multi-threaded preprocessing, parallel doc handling
RAM	64 GB DDR4 ECC	To buffer multiple document jobs
Disk	128 GB NVMe SSD	Fast random read/write for PDF/image handling
Network	1 Gbps or 10 Gbps uplink	For Groq API latency-sensitive calls
OS	Ubuntu 24.04 LTS Linux	Good CUDA/PyTorch compatibility